

INTERNSHIP REPORT ON ARTIFICIAL INTELLIGENCE AND EMERGING TECHNOLOGIES

AI Discussions • Cybersecurity Awareness • RAG Project Development | February 2026 – March 2026

Internship Organization:

Bits N' Bytes Cybersecurity Education, USA

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Introduction

This report presents an overview of my learning experiences, activities, and contributions during the internship period. The internship provided valuable exposure to important topics such as **Artificial Intelligence applications, cybersecurity awareness, ethical considerations in AI, and emerging technological trends**. Through active participation in discussions, assignments, and knowledge-sharing sessions, I gained a broader understanding of how AI technologies are influencing various sectors and shaping modern society.

In addition to participating in these learning activities, I also developed an **AI-based project titled “Advanced RAG-Enabled Transformer Embedding System for Intelligent Knowledge Discovery from Excel Datasets.”** The project enables users to interact with Excel datasets through natural language queries by utilizing **Retrieval-Augmented Generation (RAG), transformer-based embeddings, and vector search techniques** to retrieve and generate accurate insights from structured data.

The following sections summarize the **assignments completed, key learning outcomes, major contributions during the internship, and overall reflections on the experience.**

Activity Participation and Responses

Activity 1: Emergency Call Scenario and AI-Based Voice Scams

Discussion Topic:

Interns were asked how they would respond if they received a suspicious emergency call from someone claiming to be a family member.

My Response:

I would quickly take a screenshot of the call and message, then pause before reacting. Instead of panicking, I would verify the situation through our family group or confirm whether the situation matches their usual routine before believing it is real.

Learning Outcome:

This discussion highlighted the importance of **verifying emergency communications**, especially as AI voice cloning scams are becoming more common.

Activity 2: AI Impact on Computer-Based Jobs

Discussion Topic:

An article discussed how AI agents may transform computer-based jobs and potentially disrupt employment.

My Response:

AI will definitely change many computer-based jobs, but it should be viewed as a tool rather than a threat. While some traditional roles may evolve or disappear, new opportunities will also emerge for people who learn to work with AI. Students and interns should focus on practical skills, adaptability, and understanding AI tools to stay relevant in the rapidly changing technology landscape.

Learning Outcome:

AI will reshape the job market, but individuals who continuously upgrade their skills will benefit from new opportunities.

Activity 3: AI-Powered Cyber Attacks in Educational Institutions**Discussion Topic:**

Strategies for schools and colleges in India to prevent AI-driven cyber attacks.

My Response:

Indian schools must treat cybersecurity as a core institutional responsibility rather than just an IT function. With AI-driven attacks increasing, focusing on strong security fundamentals and proactive monitoring is essential to prevent serious cyber threats.

Learning Outcome:

Educational institutions must strengthen cybersecurity awareness, infrastructure, and training to protect students and digital systems.

Activity 4: Cybersecurity Organizations and Certifications**Discussion Topic:**

Government institutions and certifications that help improve cybersecurity in academic environments.

My Response:

In India, organizations like **CERT-In** and **NCIIPC** work toward strengthening cybersecurity, along with advocacy groups like the **CyberPeace Foundation**. Globally, institutions such as **NIST** and **ISC²** set important cybersecurity standards. Certifications like **CISSP**, **CISA**, **CEH**, and **CompTIA Security+** can help institutions create safer digital environments.

Learning Outcome:

Cybersecurity requires collaboration between **government institutions, global standards organizations, and trained professionals**.

Activity 5: AI Disruption in Cybersecurity Industry**Discussion Topic:**

Impact of Anthropic's Claude Code Security and other AI tools on cybersecurity markets.

My Response (Summary of Report):

- Market reactions were driven more by fear than immediate technological disruption.
- AI tools like Claude Code Security help developers detect issues early in the coding stage.

- These tools do not replace full cybersecurity systems.
- Some code-checking companies may face competition.
- Core cybersecurity functions such as ransomware protection remain essential.
- Indian IT companies may benefit if they integrate AI into their services.

Learning Outcome:

AI is transforming cybersecurity workflows but **human expertise and advanced security infrastructure remain critical.**

Activity 6: AI Deepfakes and Legal Risks

Discussion Topic:

Examples of AI misuse reported in the news.

My Response:

I shared a news article where the **Delhi High Court ordered the removal of unauthorized AI-generated deepfakes using yoga guru Swami Ramdev's persona.** Such deepfakes can mislead the public and demonstrate the risks associated with AI-generated misinformation.

Learning Outcome:

AI technologies require **legal regulation and responsible use** to prevent misinformation and identity misuse.

Activity 7: AI in National Security

Discussion Topic:

The role of AI in national security decision-making.

My Response:

AI should mainly support data analysis and cybersecurity operations, but final national security decisions must remain under human control. Strong regulations, ethical safeguards, and human oversight are essential for powerful AI systems.

Learning Outcome:

AI can assist governments in security analysis, but **human judgment must remain central to critical decisions.**

Activity 8: AI Ethics and Autonomous Weapons

Discussion Topic:

Anthropic CEO's interview discussing AI risks in military technologies.

My Response:

The discussion highlights how powerful AI systems can influence military operations. Autonomous weapons without human supervision could lead to dangerous situations. Therefore, developers must prioritize ethics, safety, and strong human oversight when building AI systems.

Learning Outcome:

AI development must always consider **ethical responsibility, human safety, and global security implications.**

Activity 9: Challenges in Technology Job Market

Discussion Topic:

How interns can prepare for careers in an AI-disrupted job market.

My Response:

The biggest challenge today is the gap between academic knowledge and industry-ready skills. Interns should focus on hands-on projects, learning AI and machine learning tools, and strengthening problem-solving abilities.

Learning Outcome:

Employers value **practical experience, adaptability, and technical skills**.

Activity 10: Increasing Use of Large Language Models (LLMs)

Discussion Topic:

The growing popularity of AI tools like ChatGPT and Claude.

My Response:

The recent events increased visibility for Anthropic and its AI tools. Personally, my usage of LLMs has increased significantly compared to a few months ago. I now use them regularly for learning, generating ideas, and assisting with tasks.

Learning Outcome:

LLMs are becoming an **important productivity and research tool across industries**.

Activity 11: AI in Sports Analytics (Cricket)

Discussion Topic:

Role of AI in sports events like the T20 Cricket Finals.

Suggested Response:

AI plays a significant role in modern cricket through real-time analytics, player performance analysis, predictive statistics, and broadcast graphics. Technologies such as ball tracking, decision review systems (DRS), and performance analytics help teams and broadcasters analyse the game more accurately than traditional manual methods.

Learning Outcome:

AI is transforming sports by enabling **real-time data analysis and strategic decision-making**.

Activity 12: AI-Based Scams in India

Discussion Topic:

Common AI scams and how people can protect themselves.

My Response:

I have seen fake job offers on LinkedIn and WhatsApp where scammers ask for registration or training fees. I avoided one by verifying the company through its official website, which confirmed it was a scam.

Learning Outcome:

Awareness and verification are critical to preventing **AI-enabled scams and online fraud**.

Activity 13: Phishing Awareness – Suspicious Courier Message

Discussion Topic:

A message claiming to be from a postal service asked to check an attachment regarding a package.

My Response:

This appears to be a **phishing message**. It is unsafe to open attachments from unknown numbers, especially without proper tracking details. Instead, the package status should be verified through the **official courier website**.

Learning Outcome:

This activity highlighted the importance of **avoiding suspicious links or attachments**, verifying information through trusted sources, and staying alert to phishing attempts.

Activity 14: Advanced RAG-Enabled System for Excel Dataset Knowledge Discovery

Assignment / Project Topic:

During the internship, I developed a project titled “**Advanced RAG-Enabled Transformer Embedding System for Intelligent Knowledge Discovery from Excel Datasets.**” The objective of this project was to design an AI-driven system that allows users to analyze and interact with Excel datasets using natural language queries.

Through continuous support and guidance from the trainer, the project focused on implementing modern AI techniques to make structured data more accessible and easier to interpret. This experience helped me apply theoretical knowledge to a practical application while exploring advanced concepts in **Artificial Intelligence and data analysis**.

Project Description:

The system enables users to upload Excel datasets and ask questions about the data in natural language. It leverages **transformer-based embeddings** to convert rows of the dataset into semantic vectors. These vectors are stored in a **FAISS vector database**, which allows efficient similarity-based retrieval of relevant information.

The project integrates a **Retrieval-Augmented Generation (RAG) pipeline**, where the most relevant dataset rows retrieved from the vector database are provided as contextual input to a **Large Language Model (LLM)**. The LLM then generates accurate and context-aware responses based on the retrieved data.

Technical Implementation:

- Excel datasets are uploaded and processed for structured analysis.
- Dataset rows are converted into **semantic embeddings using transformer models**.
- Embeddings are stored in a **FAISS vector database** for efficient similarity search.
- When a user asks a query, the system retrieves the most relevant dataset rows using **vector similarity matching**.
- The retrieved information is passed to an **LLM through a RAG pipeline** to generate grounded answers.

- The system includes a **REST API built with FastAPI** to handle backend operations.
- An **interactive dashboard developed using React** allows users to upload datasets, query the system, and visualize insights.

Learning Outcomes:

Through this project, I gained practical experience in:

- Designing **AI-powered data analysis systems**
- Implementing **semantic search using transformer embeddings**
- Building **Retrieval-Augmented Generation (RAG) pipelines**
- Working with **vector databases such as FAISS**
- Developing **REST APIs using FastAPI**
- Creating **interactive dashboards using React** for monitoring AI performance and visualizing results
- Understanding how AI can enable **intelligent knowledge discovery from structured datasets**

This project strengthened my understanding of **AI application development, modern NLP architectures, and real-world data analysis systems.**

Summary of What I Have Learned So Far

Through this internship, I gained valuable knowledge in multiple areas related to Artificial Intelligence and technology applications. Some of the key learnings include:

- Understanding the **real-world impact of AI in cybersecurity, employment, national security, and digital safety.**
- Learning about **ethical concerns and risks associated with AI systems such as deepfakes, voice cloning, and autonomous weapons.**
- Improving awareness of **cybersecurity practices and institutions responsible for protecting digital systems.**
- Understanding the **importance of adaptability and continuous learning in the AI-driven job market.**
- Developing practical experience in **AI application development, especially using RAG architectures and vector databases.**

Overall, the internship helped me understand that AI is not only a technical field but also has **social, ethical, and economic implications.**

Experiences or Opportunities I Would Have Liked to Gain

While the internship has been very informative, I would have also appreciated:

- More **hands-on collaborative projects with other interns.**

- Additional **technical workshops on advanced AI tools and real-world AI deployment.**
- Industry expert sessions focusing on **AI careers and industry expectations.**

These experiences could further enhance practical learning and industry readiness.

Willingness to Volunteer for BitsNBytes Chapter

Yes, I would be willing to volunteer for the **BitsNBytes chapter at Panimalar.** Participating in organizing events and helping students explore AI and technology would be a valuable opportunity to contribute to the learning community and gain leadership experience. I would be open to contributing approximately **6 hours per week** if required.

Comparison of LLMs and AI Tools Used

During the internship, I frequently used several AI tools for research, analysis, and project development.

ChatGPT

- Used for research, explanations, and coding assistance.
- Helpful for brainstorming ideas and generating structured documentation.

Claude (Anthropic)

- Effective for longer analytical responses and summarizing complex information.

Google Gemini

- Useful for quick searches and integration with other Google services.

Overall, each tool has strengths. I generally use **ChatGPT for technical guidance and structured responses**, **Claude for detailed analysis**, and **Gemini for quick information retrieval.**

Need for Additional Career Mentoring

Yes, additional career mentoring would be beneficial, especially in the following areas:

- Understanding **AI and data science career pathways.**
- Guidance on **building industry-ready projects.**
- Advice on **internships, research opportunities, and job preparation.**

Such mentoring would help bridge the gap between academic learning and industry expectations.

My Best Contribution During the Internship

I believe my best contribution during the internship was developing the **Advanced RAG-Enabled Transformer Embedding System for Intelligent Knowledge Discovery from Excel Datasets**.

This project demonstrates how AI can be used to interact with structured datasets using natural language queries and semantic search techniques. It integrates **transformer embeddings, FAISS vector databases, RAG pipelines, FastAPI backend services, and a React-based dashboard**.

This project helped showcase the practical implementation of modern AI architectures.

Areas Where I Could Improve

One area where I could improve is **participating even more actively in collaborative discussions with other interns** and exploring additional experimental ideas beyond the assigned tasks. With more time, I would also expand the project with **more datasets and advanced analytics features**.

Communication with Trainer or Team

During the internship, communication occurred through:

- **Group messages and discussion forums**
- **Online meetings and mentoring sessions**
- **Direct messages for clarification and feedback**

Throughout the internship, I maintained regular communication with the trainer and team members several times each week through messages, discussions, and meeting updates for guidance, feedback, and progress updates.

Recommendation of the Internship

Yes, I would highly recommend this internship to other students in my college. The program offers valuable exposure to **Artificial Intelligence technologies, cybersecurity awareness, ethical considerations, and real-world technology trends**. It encourages critical thinking and helps students develop a deeper understanding of the broader impact of AI beyond academic knowledge. Overall, it is a meaningful learning experience that supports both **technical growth and awareness of real-world applications**.

My Reflections on the Internship Experience

Overall, the internship has been a highly meaningful and enriching learning experience. I truly enjoyed the balanced approach of the program, which combined both **technical exploration and broader discussions on the societal impact of Artificial Intelligence**. The technical components helped me strengthen my understanding of modern AI concepts, tools, and practical applications, while the non-technical discussions encouraged critical thinking about **AI ethics, cybersecurity awareness, digital responsibility, and the broader implications of emerging technologies**.

Participating in thoughtful discussions, completing assignments, and developing an AI-based project significantly enhanced my **analytical thinking, problem-solving abilities, and practical understanding of AI systems**. At the same time, the internship also emphasized the importance of **ethical responsibility, security awareness, and responsible innovation** in the development and use of AI technologies.

Overall, the experience has been both **educational and inspiring**, helping me develop a deeper appreciation for how Artificial Intelligence can be applied responsibly to create meaningful and impactful solutions.

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